Statistical Arbitrage via Option Volatility Skew

MS&E 244 Project

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Outline

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Strategy Overview

Core Idea

Use option-implied volatility skew as a cointegrated signal between individual stocks and their sector ETFs.

1. Compute the **volatility surface** from option data for each ticker
2. Extract the **implied volatility skew** as a scalar time series
3. Model the **cointegration** between stock skew and sector ETF skew
4. Trade the **mean-reverting spread** between the two
5. **Delta-hedge** to remove exposure to the underlying equity

Data & Cleaning

Raw Data

- $\sim$10 GB of option data
- Sample period: **2015–2019**

Filters Applied

- Remove options with < 1 week to expiration
  (noisy near expiry)
- Remove options with implied volatility $>$ threshold
  (some values exceeded 800%)

Why These Filters?

Very short-dated options exhibit large bid-ask spreads and gamma blow-ups that distort the skew estimate. Extreme IV values are data errors or illiquid deep-OTM strikes.

Step 1 — Constructing the Skew

For each day and each ticker, fix a target τ^* and $\Delta^* = 0.25$:

1. Use **OTM options only** (higher liquidity; avoids early-exercise noise)
2. Select the expiry nearest to τ^*
3. Find the OTM put and OTM call whose $|\Delta|$ is closest to Δ^*
4. Define the skew as the **25-delta risk-reversal IV spread**:

$$\text{skew}_t \;=\; \sigma^{\text{put}}_{25\Delta,\,t} \;-\; \sigma^{\text{call}}_{25\Delta,\,t}$$

Alternatives: polynomial fit & simple

Fit $\sigma_{IV}(\kappa) \approx \beta_0 + \beta_1\kappa + \beta_2\kappa^2$ to all OTM options and use $-\hat{\beta}_1$ as the skew proxy (negated so that high = puts expensive). Smoother but less directly tied to the 25Δ strikes actually traded. Can also just use the endpoints of the smile.

Step 2 — Trading the Spread

We now have skew time series $\{s_t^{\text{stock}}\}$ and $\{s_t^{\text{sector}}\}$.

1. **Cointegration regression**

$$s_t^{\text{stock}} = \alpha + \beta\, s_t^{\text{sector}} + \varepsilon_t$$

2. **Rolling z-score** of residuals ε_t

$$z_t = \frac{\varepsilon_t - \mu_\varepsilon}{\sigma_\varepsilon}$$

3. **Entry / exit rules** (mean-reversion):
   - Enter when $|z_t| > z_{\text{entry}}$
   - Exit when $|z_t| < z_{\text{exit}}$
4. **Position sizing**: if spread is high, go long stock-skew reversion and short sector-skew reversion, weighted by $\hat{\beta}$

Step 3 — Delta Hedging

Why Delta Hedge?

A pure options spread position carries first-order equity exposure. We neutralise this to isolate the *skew* signal.

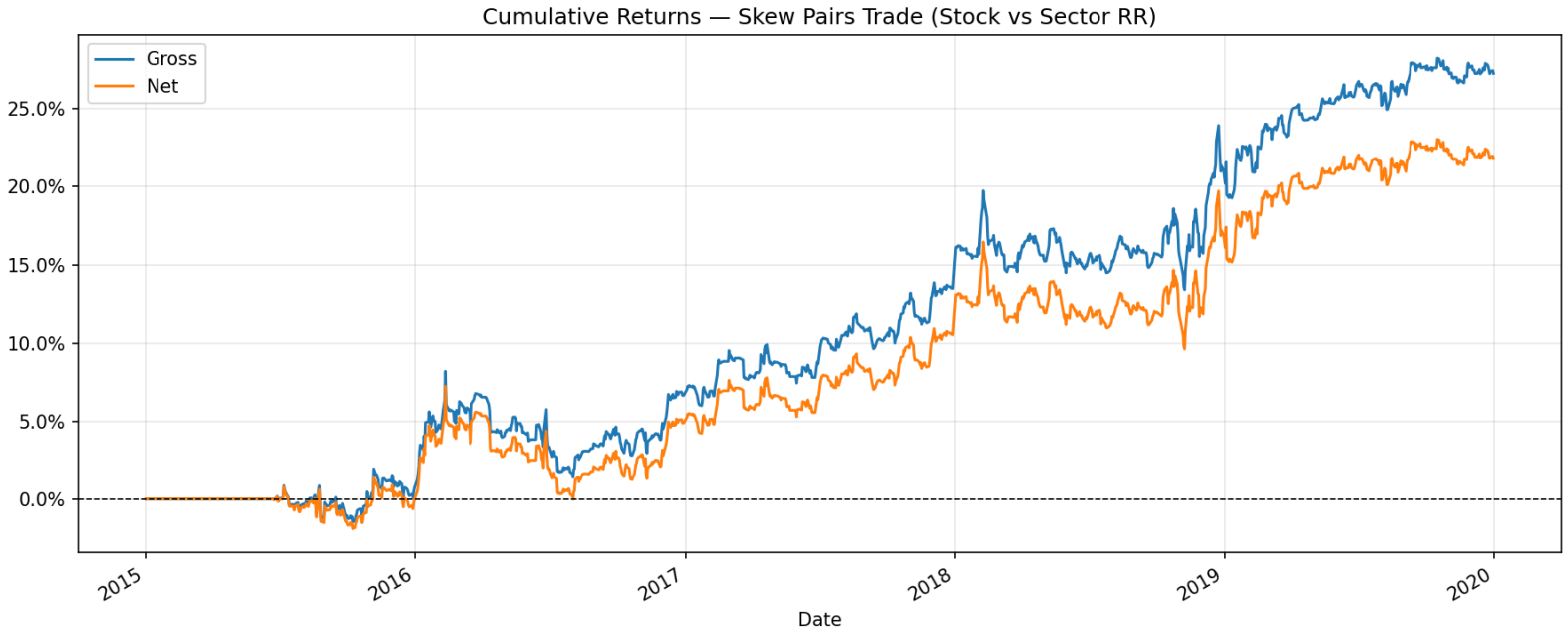
- Compute Black-Scholes Δ for each option leg
- Trade the underlying stock / ETF to flatten the net delta daily
- This converts the strategy into a **volatility / skew trade**, independent of equity market direction

Overall stats

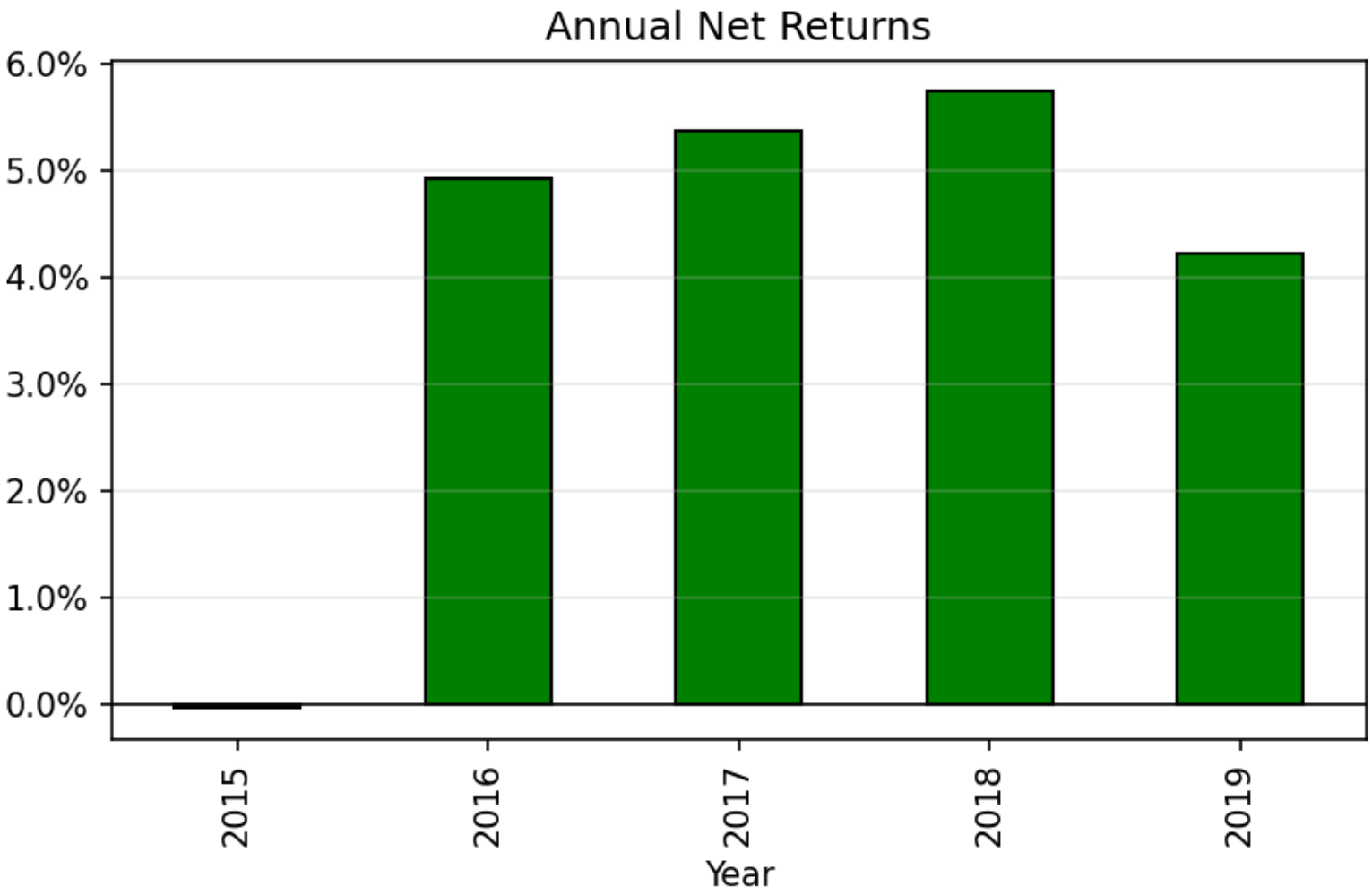
Table: Portfolio Performance Summary

Metric	**Value**
Gross Total Return	27.28%
Gross Ann. Return	4.95%
Gross Ann. Volatility	5.40%
Gross Sharpe Ratio	0.92
Gross Max Drawdown	-6.29%
Net Total Return	21.80%
Net Ann. Return	4.03%
Net Ann. Volatility	5.40%
Net Sharpe Ratio	0.75
Net Max Drawdown	-6.78%
Total Transaction Cost	4.40%

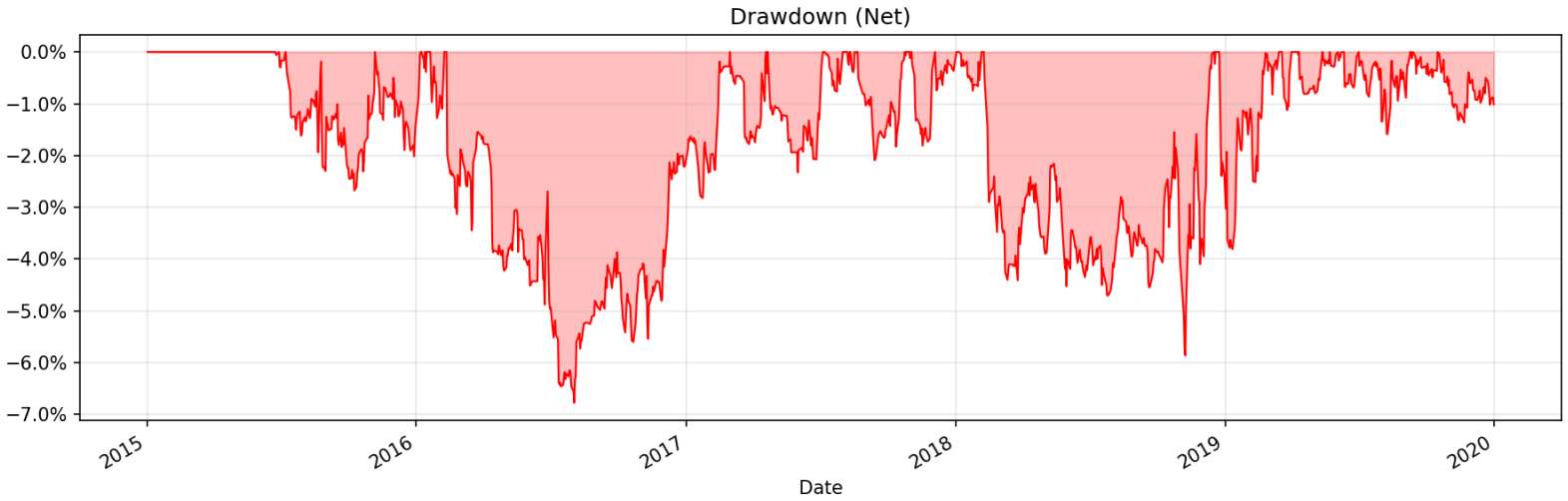
Cumulative Returns



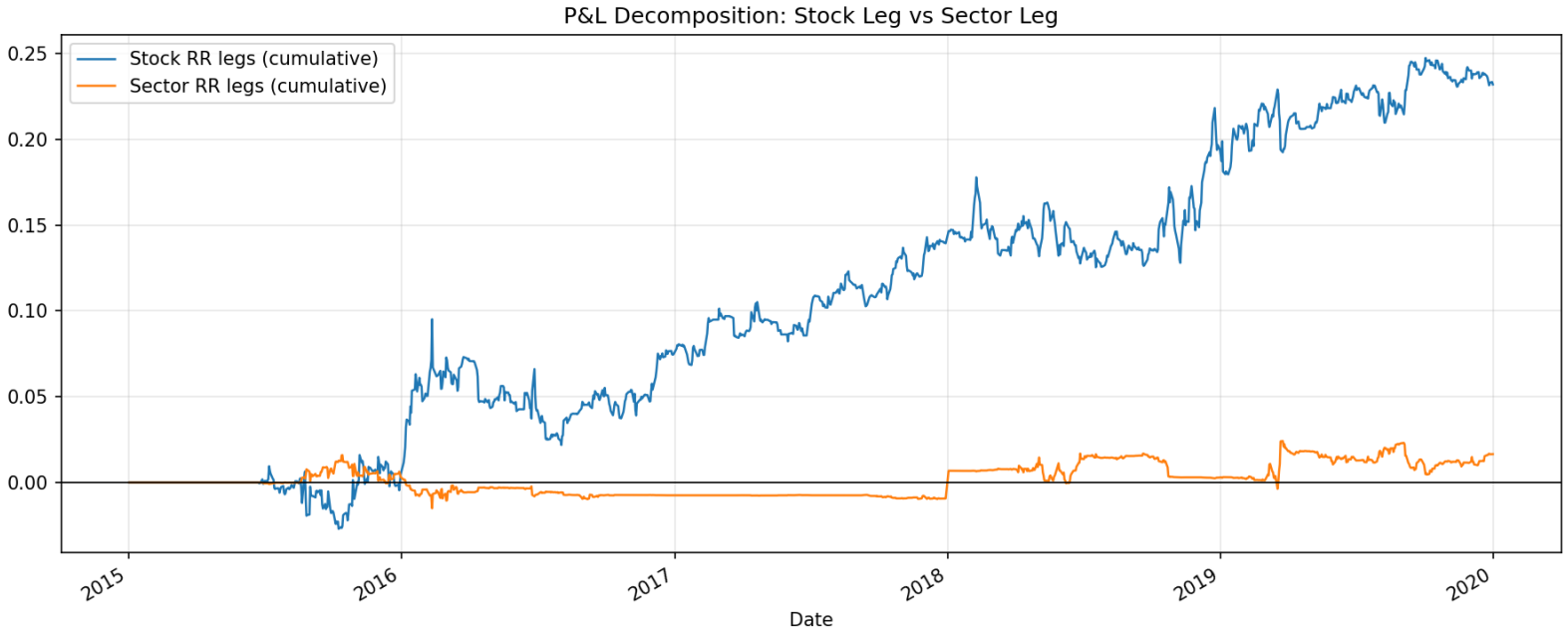
Annual Returns



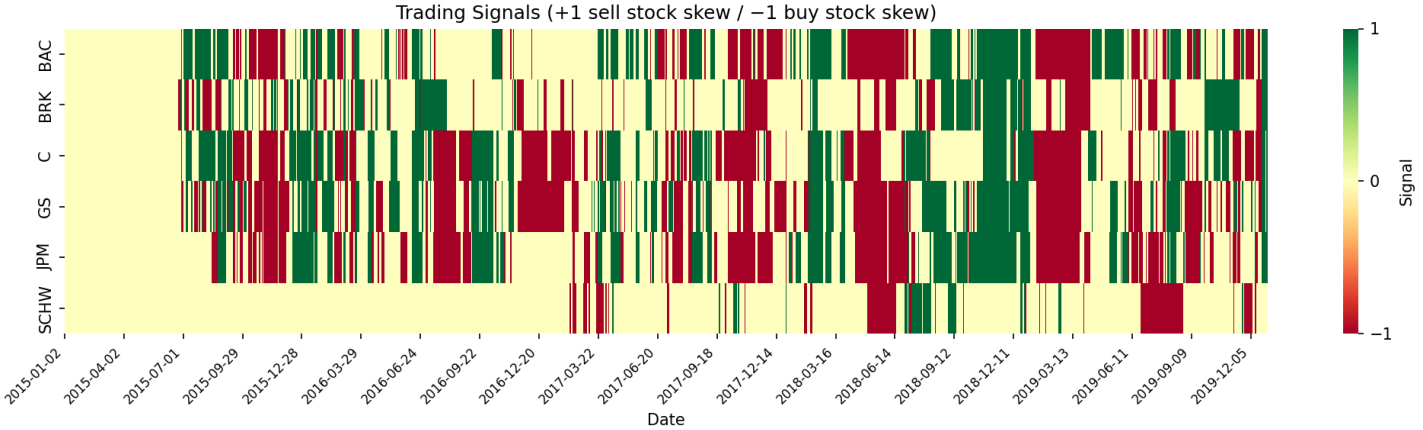
Drawdown


Drawdown (Net)
0.0%
−1.0%
−2.0%
−3.0%
−4.0%
−5.0%
−6.0%
−7.0%
2015
2016
2017
2018
2019
2020
Date

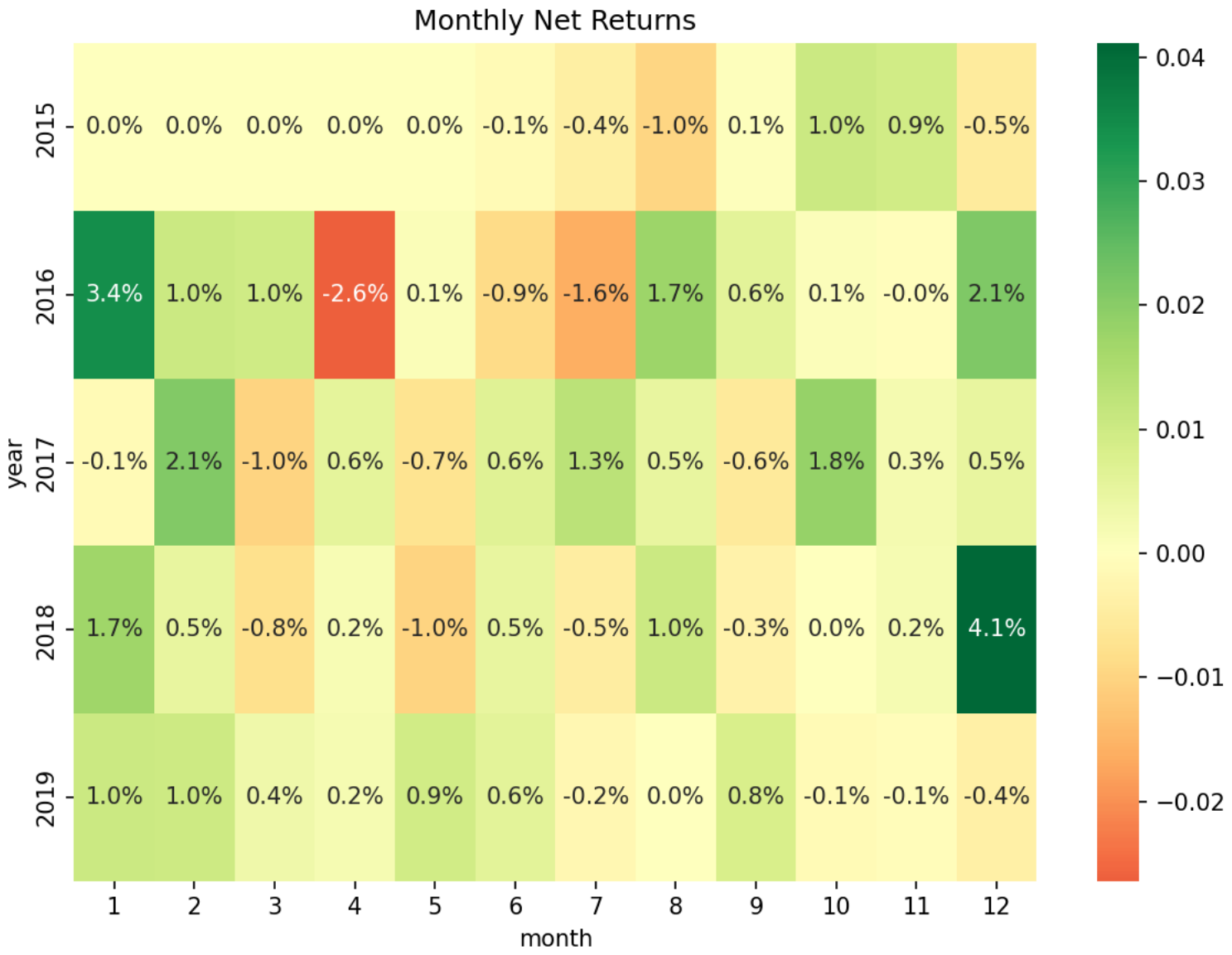
P&L Decomposition



Signals


Trading Signals (+1 sell stock skew / −1 buy stock skew)
BAC
BRK
C
GS
JPM
SCHW
2015-01-02
2015-04-02
2015-07-01
2015-09-29
2015-12-28
2016-03-29
2016-06-24
2016-09-22
2016-12-20
2017-03-22
2017-06-20
2017-09-18
2017-12-14
2018-03-16
2018-06-14
2018-09-12
2018-12-11
2019-03-13
2019-06-11
2019-09-09
2019-12-05
Date
1
0
−1
Signal

Monthly Returns Heatmap



Conclusion & Future Work

Summary

- Extracted skew as the 25Δ put–call IV spread
- Traded cointegrated stock–sector skew spread
- Delta-hedged to isolate skew signal

Future Extensions

- Use full volatility surface (not just one τ)
- Weight by open interest
- SVI / signature-method skew parametrisation (beyond polynomial fit)
- Richer cointegration models (Engle-Granger, Johansen)

Key Takeaway

Option skew cointegration provides a market-neutral, delta-hedged signal with meaningful mean-reversion properties.

Thank you!

Questions?